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Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Core Course (CC) - MG1CCRBBA102 - BUSINESS STATISTICS AND LOGIC

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** out of **8** Questions

Each Question carries **2** marks

1. Define the Harmonic Mean and discuss its applicability. [K] / [CO1]
2. What is bar diagram? [K] / [CO1, CO2]
3. Define simple regression. [K] / [CO5]
4. The coefficient of correlation between customer satisfaction scores and sales is 0.68 and the value of probable error is 0.1. How do you interpret the correlation? [U] / [CO5]
5. What is skewness? [K] / [CO1]
6. Illustrate with example the term "equally likely events". [A] / [CO3]
7. What is meant by progression? [K] / [CO2]
8. Determine the relationship between two individuals given that one is the grandfather of the other. [An] / [CO2]

[5x2 = 10]

Part B

Short Essay Type Questions

Answer any **4** out of **6** Questions

Each Question carries **5** marks

9. Compute the quartile deviation and inter-quartile range of the following series [A] / [CO4]

Age Group	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100
No. of Persons	4	10	15	20	11

10. What is Arithmetic Mean? A company tracks the number of new customer registrations at two of its locations over 5 weeks. [An] / [CO3, CO4]
    New Registrations in Location A: 50, 45, 55, 60, 52
    New Registrations in Location B: 40, 48, 45, 50, 55
    Calculate the average number of new customer registrations for each location. Interpret the results.

11. Explain how to calculate range when the given data is continuous. Also find range and coefficient of range of the following series. [A] / [CO4]

Marks	**20 - 30**	**30 - 40**	**40 - 50**	**50 - 60**	**60 - 70**
No. of Students	8	12	20	7	3

12. Given a dataset of student test scores, create a frequency distribution table. Use appropriate class intervals (40-50, 50-60, 70-80, 80-90), calculate class frequencies, and define the class limits for each interval. Dataset of student test scores: 45, 60, 78, 89, 56, 67, 73, 81, 92, 58, 64, 85, 49, 77, 68 [A] / [CO2]

13. A housewife uses 10 kg wheat, 20 kg fuel, 5 kg sugar and 2 kg oil. Prices per kg of these items are respectively Rs 1.50, Re 0.50, Rs 2.80 and Rs 10. Taking quantities used as weights find out the weighted arithmetic average of the prices. [A] / [CO4]

14. The probability that there is one error in an account statement prepared by A is 0.3 and for B and C, they are 0.4 and 0.45 respectively. A , B and C prepared 20,10 and 40 statements respectively. Find the expected number of errors in all. [A] / [CO3]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **2** out of **4** Questions

Each Question carries **20** marks

15. Calculate standard deviation and variance for the following data [A] / [CO4]

Class (Range)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90
Frequency	3	8	12	20	30	15	7	5	0

16. Calculate mean deviation about median for the following data. Also, find the mean deviation about the coefficient of mean deviation. [A] / [CO4]

x (Range)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
f (Frequency)	18	16	15	12	10	5	2	2

17. Given the following data , calculate [A] / [CO5]
    (i) The probable value of Y When X = 12
    (ii) The probable value of X when Y =30

	x	y
Mean	27.6	14.8
S.D	40	20

    r =0.8

18. (a) Define scatter diagram . How is it used in business. (b) Using scatter diagram how will you ascertain: i) perfect correlation ii) No correlation iii) Limited degree of correlation . In each case draw the rough sketch of the graph. [U] / [CO5]

[2x20 = 40]